

WILMS

Women's Interest-Free Loan Management System

Notification & Communication Book — v1.8.0

WILMS Notification & Communication Book

Version: 1.7.3 Classification: Confidential

1. Overview

WILMS delivers multi-channel notifications and a Communications Center for programme-wide messaging.

2. Notification channels

| Channel | Use cases |

|-----|-----|

| In-app | All event types; notification inbox |

| Email | Login alerts, invitations, expense reviews |

| SMS | Borrower lifecycle (registration through completion), payment reminders, field alerts |

3. Notification guarantees

3. Notification guarantees

Deduplication

Same event within deduplication window produces single notification. Prevents alert fatigue from retry or duplicate dispatch.

Quiet hours

Organisation-configured quiet hours suppress non-critical notifications. Critical security alerts (login from new device) may override.

Cron dispatch

Daily batch dispatch at 06:00 UTC via Vercel Cron (/api/cron/notifications). Failed dispatches logged; retry on next cycle.

4. Event types

| Event | Channels | Recipients |

|-----|-----|-----|

| Registration submitted / approved | SMS, email (optional/yes), in-app | Borrower (SMS), officer/collector |

| Loan created / approved | SMS, email, in-app | Borrower (SMS), collector |

| Admin fee / disbursement / schedule | SMS, email, in-app | Borrower (SMS), collector |

| Payment reminder / due today | SMS; email optional or none | Borrower |

| Missed / grace / escalation | SMS; email per matrix | Borrower, collector, Super Admin |

| Payment received / loan completed | SMS, email, in-app | Borrower (SMS), collector |

| Login alert | Email | User |

| Invitation accepted | In-app | Admin |

| Expense submitted | In-app, Email | Reviewers |

| Expense approved/rejected | In-app, Email | Submitter |

| Reconciliation variance | In-app | Admin, Approver |

| Overpayment review | In-app | Approver |

| Ops incident | In-app | Admin |

| Maintenance window | In-app | All users |

Borrower SMS copy and timing: documentation/notifications/.

5. In-app notification inbox

Accessible from navbar bell icon. Shows unread count. Mark individual or all as read. Role-aware filtering — users see only relevant notifications.

6. Communications Center

Templates

Reusable message templates with variable substitution (e.g., {borrowerName}, {amount}). Admin creates and manages templates.

Broadcasts

Targeted messages to role groups or all users. One-way communication from HQ to field.

Message threads

Two-way messaging between HQ staff and field collectors. Threaded conversation view.

Audit

All sent messages logged with sender, recipients, timestamp, and content hash.

7. Configuration

Super Admin ! Settings:

- Notification preferences per channel
- Quiet hours schedule

- Email/SMS provider credentials (environment variables)

8. API endpoints

Method	Path	Description
GET	/notifications	List user notifications
PATCH	/notifications/:id/read	Mark read
POST	/communications/broadcasts	Send broadcast
GET	/messages	Message threads

9. Planned enhancements (v1.9+)

- Scheduled report delivery via notification
- Advanced routing rules
- Borrower SMS reminders (v2.5)
- Push notifications via PWA service worker

CONFIDENTIAL — For authorized WILMS personnel, executive review, procurement, and official record keeping only.